

Step 8: Launching an EC2 instance

Now that you have configured your EC2 instance settings, it is time to launch your instance.

21. In the right pane, choose **Launch instance**

22. Choose **View all instances**

The instance appears in a **Pending** state, which means it is being launched. It then changes to **Running**, which indicates that the instance has started booting. There will be a short time before you can access the instance.

The instance receives a public DNS name that you can use to contact the instance from the Internet.

23. Select the ☐ box next to your **Web Server**. The **Details** tab displays detailed information about your instance.

💬 To view more information in the **Details** tab, drag the window divider upward.

Review the information displayed in the **Details**, **Security** and **Networking** tabs.

24. Wait for your instance to display the following:

Note: Refresh if needed.

- **Instance State:** ● Running
- **Status Checks:** ✔ 2/2 checks passed

Task 2: Monitor Your Instance

◦ **Status Checks:** ✔ 2/2 checks passed

Task 2: Monitor Your Instance

Monitoring is an important part of maintaining the reliability, availability, and performance of your Amazon Elastic Compute Cloud (Amazon EC2) instances and your AWS solutions.

25. Select the instance by checking the box next to the instance and navigate to the bottom of the screen to the **Status checks** tab.

❗ With instance status monitoring, you can quickly determine whether Amazon EC2 has detected any problems that might prevent your instances from running applications. Amazon EC2 performs automated checks on every running EC2 instance to identify hardware and software issues.

Notice that both the **System reachability** and **Instance reachability** checks have passed.

26. Select the **Monitoring** tab.

This tab displays Amazon CloudWatch metrics for your instance. Currently, there are not many metrics to display because the instance was recently launched.

You can choose a graph to see an expanded view.

❗ Amazon EC2 sends metrics to Amazon CloudWatch for your EC2 instances. Basic (five-minute) monitoring is enabled by default. You can enable detailed (one-minute) monitoring.

27. In the **Actions** menu, select **Monitor and troubleshoot** ▶ **Get Instance Screenshot**.

Task 1: Launching your EC2 instance

In this task, you will launch an Amazon EC2 instance with *termination protection*. Termination protection prevents you from accidentally terminating an EC2 instance. You will deploy your instance with a User Data script that will allow you to deploy a simple web server.

5. In the AWS Management Console on the **Services** menu, choose **EC2**.
6. In the left navigation pane, choose **EC2 Dashboard** to ensure that you are on the dashboard page.
7. Choose **Launch instance**, and then select **Launch instance**.

Step 1: Naming your EC2 instance

When you name your instance, AWS creates a key value pair. The key for this pair is **Name**, and the value is the name you enter for your EC2 instance.

8. In the **Name and tags** pane, in the **Name** text box, enter `Web Server`.

Step 2: Choosing an Amazon Machine Image (AMI)

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Step 2: Choosing an Amazon Machine Image (AMI)

An AMI provides the information required to launch an instance, which is a virtual server in the cloud. An AMI includes the following:

- A template for the root volume for the instance (for example, an operating system or an application server with applications)
- Launch permissions that control which AWS accounts can use the AMI to launch instances
- A block device mapping that specifies the volumes to attach to the instance when it is launched

The **Quick Start** list contains the most commonly used AMIs. You can also create your own AMI or select an AMI from the AWS Marketplace, an online store where you can sell or buy software that runs

Keep this setting.

Step 3: Choosing an instance type

Amazon EC2 provides a wide selection of *instance types* optimized to fit different use cases. Instance types comprise varying combinations of CPU, memory, storage, and networking capacity and give you the flexibility to choose the appropriate mix of resources for your applications. Each instance type includes one or more *instance sizes* so that you can scale your resources to the requirements of your target workload.

Select a **t3.micro** instance. This instance type has 2 virtual CPU and 1 GiB of memory.

11. From the dropdown, select **t3.micro**.

NOTE: You may be restricted from using other instance types in this lab.

Step 4: Configuring a key pair

Amazon EC2 uses public-key cryptography to encrypt and decrypt login information. To log in to your instance, you must create a key pair, specify the name of the key pair when you launch the instance, and provide the private key when you connect to the instance.

In this lab, you do not log in to your instance, so you do not require a key pair.

12. In the **Key pair (login)** pane, select **Proceed without a key pair (Not recommended)**.

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Step 5: Configuring the network settings

You use this pane to configure networking settings.

The **VPC** indicates which virtual private cloud (VPC) you want to launch the instance into. You can have multiple VPCs, including different ones for development, testing, and production.

13. In the **Network settings** pane, choose

14. For **VPC - required**, select **Lab VPC**.

15. Still in the **Network settings** pane, configure the Security Group as follows:

- **Security group name - required:** Web Server security group
- **Description:** Security group for my web server

A *security group* acts as a virtual firewall that controls the traffic for one or more instances. When you launch an instance, you associate one or more security groups with the instance. You add *rules* to each security group that allow traffic to or from its associated instances. You can modify the rules for a security group at any time; the new rules are automatically applied to all instances that are associated with the security group.

16. Under **Inbound security groups rules** select the

In this lab, you will not log into your instance using SSH. Removing SSH access will improve the security.

Step 6: Adding storage

Amazon EC2 stores data on a network-attached virtual disk called Amazon Elastic Block Store (Amazon EBS).

You launch the EC2 instance using a default 8 GiB disk volume. This is your root volume (also known as a boot volume).

17. In the **Configure storage** pane, keep the default storage configuration.

Step 7: Configuring advanced details

18. Expand the **Advanced details** pane.

19. Select the dropdown for **Termination protection**, then choose **Enable**.

When you launch an instance in Amazon EC2, you have the option of passing user data to the instance. These commands can be used to perform common automated configuration tasks and even run scripts after the instance starts.

20. Copy the following commands, and paste them into the **User data** text box.

```
#!/bin/bash
yum -y install httpd
systemctl enable httpd
```